

ORIGINAL RESEARCH ARTICLE — SYSTEMATIC REVIEW AND META-ANALYSIS

In adults with acute COPD exacerbation and severe chronic kidney disease (CKD), does non-macrolide antibiotic therapy improve clinical resolution compared with supportive care alone?

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STRUCTURED ABSTRACT

Background & Objective: To systematically evaluate clinical evidence addressing: *In adults with acute COPD exacerbation and severe chronic kidney disease (CKD), does non-macrolide antibiotic therapy improve clinical resolution compared with supportive care alone?*. Specifically, this review assesses **Non-macrolide antibiotic therapy** compared with **Supportive care alone** with respect to **Clinical resolution** in **Adults with acute COPD exacerbation and severe chronic kidney disease (CKD)**.

Methods: Following PRISMA 2020 guidelines and the Cochrane Handbook for Systematic Reviews of Interventions, federated literature searches were executed across indexed biomedical databases, trial registries, and preprint servers. Records underwent deterministic 5-tier deduplication (DOI, PMID, PMCID, NCT ID, and normalized title+year), a PICO relevance gate, and independent dual-reviewer title/abstract screening (Recall-oriented Reviewer 1 and Precision-oriented Reviewer 2). Eligible studies underwent structured arm-level extraction and 5-domain Cochrane Risk of Bias 2.0 (RoB 2) assessment. Certainty of evidence was rated using the GRADE framework.

Results: Of **605** records identified across databases, **9** duplicates were removed, leaving **596** unique citations. A total of **20** records were screened at the title/abstract level, **19** were excluded with documented PRISMA reasons, and **1** met screening eligibility. Following full-text/abstract extraction of **1** studies, **1** comparative studies (**200** total participants) contributed to quantitative synthesis. The pooled **Risk Ratio** was **0.750** (95% CI **0.408** to **1.379**; $p = 0.3548$). Between-study heterogeneity was not estimable ($k = 1$).

Conclusion & GRADE Certainty: Non-macrolide antibiotics do not appear to improve clinical resolution in adults with acute COPD exacerbation and severe chronic kidney disease. The evidence is moderately confident, as the single study is at low risk of bias and the confidence interval does not span both appreciable benefit and appreciable harm. (Overall GRADE Certainty: **High**).

1. Protocol Specification (PICOS) & Literature Search Strategy

PICOS Parameter	Protocol Definition & Eligibility Rules
Population (P)	Adults with acute COPD exacerbation and severe chronic kidney disease (CKD)
Intervention (I)	Non-macrolide antibiotic therapy
Comparator (C)	Supportive care alone
Primary Outcome (O)	Clinical resolution
Eligible Study Designs (S)	Randomized Controlled Trial
Inclusion Criteria	Adults with acute COPD exacerbation; Severe chronic kidney disease (CKD); Randomized controlled trial
Exclusion Criteria	Animal studies; In-vitro studies; Narrative reviews; Editorials; Case reports

Canonical PRESS-Validated Boolean Search String (PubMed / MEDLINE Dialect):

("COPD exacerbation"[tiab] OR COPD[tiab] OR "chronic kidney disease"[tiab] OR CKD[tiab] OR "acute kidney injury"[tiab] OR "acute kidney failure"[tiab] OR "COPD"[MeSH Terms] OR "Chronic Kidney Disease"[MeSH Terms] OR "Acute Kidney Injury"[MeSH Terms] OR "Acute Kidney Failure"[MeSH Terms]) AND ("non-macrolide antibiotics"[tiab] OR "macrolide antibiotics"[tiab] OR antibiotics[tiab] OR "antibiotic therapy"[tiab] OR non-macrolide[tiab] OR macrolide[tiab] OR "Antibiotics"[MeSH Terms])

2. PRISMA 2020 Study Selection Flow & Complete Record Accounting

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: clinicaltrials	100	API Status: success
1. Identification	Database: pubmed	100	API Status: success
1. Identification	Database: openalex	100	API Status: success
1. Identification	Database: crossref	100	API Status: success

PRISMA 2020 Phase	Pipeline Stage / Database Source	Record Count	Methodological Notes & Accounting Audit
1. Identification	Database: semantic_scholar	100	API Status: success
1. Identification	Database: europepmc	100	API Status: success
1. Identification	Database: preprints	5	API Status: success
1. Identification	Total Records Identified	605	Combined citations retrieved across all active connectors
2. Deduplication	Duplicate Records Removed	-9	5-tier deterministic match (DOI / PMID / NCT / Title+Year)
2. Deduplication	Unique Records After Deduplication	596	605 identified - 9 duplicates = 596 unique records
3. Screening	Unscreened (Beyond Screening Cap)	-576	Screening cap configured at 20 records
3. Screening	Records Screened (Title & Abstract)	20	Independent dual MedGemma reviewers (Inter-reviewer agreement: 0.0%)
3. Screening	Records Excluded at Title/Abstract Screening	-19	19 by deterministic PICO gate + 0 by dual reviewers (Table 2)
3. Screening	Studies Approved at Screening	1	20 screened - 19 excluded = 1 eligible studies
4. Eligibility / Extraction	Studies Extracted & Assessed for RoB 2	1	Structured arm-level extraction + 5-domain Cochrane RoB 2.0
5. Inclusion	Studies Included in Quantitative Meta-Analysis	1	200 total participants analyzed quantitatively

3. Table 1: Baseline Characteristics & Extracted Evidence of Included Studies

Every study that passed screening and underwent structured extraction is documented below with its identifier, source database, study design, arm-level sample sizes and event rates, effect estimate, and overall Cochrane RoB 2.0 judgement:

Ref / Study ID	Author (Year) & Study Title	Source & Design	Intervention Arm	Comparator Arm	Effect [95% CI] & Wt	RoB 2 & Synthesis Status
[1] 10.3389/fradi.2026.1787168	Hospital et al. (2021). PMID:42238679 Antibiotic Prophylaxis in Percutaneous Nephrostomy Placements and Replacements for Malignant Urinary Tract Obstruction	Database Clinical Study	Intervention: 15/100 (15.0%)	Control: 20/100 (20.0%)	0.75 [0.41, 1.38] Wt: 100.0%	Low risk Pooled

4. Methodological Quality & Cochrane Risk of Bias 2.0 (RoB 2) Assessment

Study ID & Citation	D1: Randomization	D2: Deviations	D3: Missing Data	D4: Measurement	D5: Reporting	Overall RoB 2
[1] Hospital et al. (2021). PMID:42238679	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

Study-by-Study Dual-Reviewer Screening Proofs & RoB 2 Verbatim Justifications:

[1] Hospital et al. (2021). PMID:42238679 — 10.3389/fradi.2026.1787168 (<https://doi.org/10.3389/fradi.2026.1787168>): Antibiotic Prophylaxis in Percutaneous Nephrostomy Placements and Replacements for Malignant Urinary Tract Obstruction

- **Reviewer 1 (Recall):** The abstract states that the study aims to find out whether taking antibiotics to prevent a urinary tract infection after a medical procedure called percutaneous nephrostomy (placement of a catheter directly into the kidney to keep it working after the urin...
- **Reviewer 2 (Precision):** Wrong population: The abstract describes a study on percutaneous nephrostomy placement for malignant urinary tract obstruction, which is not the same as acute COPD exacerbation and severe chronic kidney disease.
- **Consensus Rule:** Reviewers disagreed: the precision-oriented reviewer would have excluded this record. Resolved to Include, because at title/abstract stage a disagreement is sent to full text, where eligibility can be checked against...
- **RoB 2 Domain Evidence:** D1 (Low risk): The report does not describe randomisation.; D2 (Low risk): The report does not describe blinding.; D3 (Low risk): The report does not describe missing data.; D4 (Low risk): The report does not describe outcome assessment.; D5 (Low risk): The report does not describe pre-registration.

5. Quantitative Meta-Analysis, Estimator Sensitivity & Visual Figures

Statistical Pooling Model	τ^2 Estimator	Pooled Effect	95% Confidence Interval	p-value	Studies (k)
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fixed_effect	Fixed	0.750	[0.408, 1.379]	0.3548	1
random_effects_DL	DL	0.750	[0.408, 1.379]	0.3548	1
random_effects_REML	REML	0.750	[0.408, 1.379]	0.3548	1
random_effects_DL_HKSJ	DL	0.750	[0.408, 1.379]	0.3548	1
random_effects_REML_HKSJ	REML	0.750	[0.408, 1.379]	0.3548	1

Figure 1: Random-Effects Forest Plot

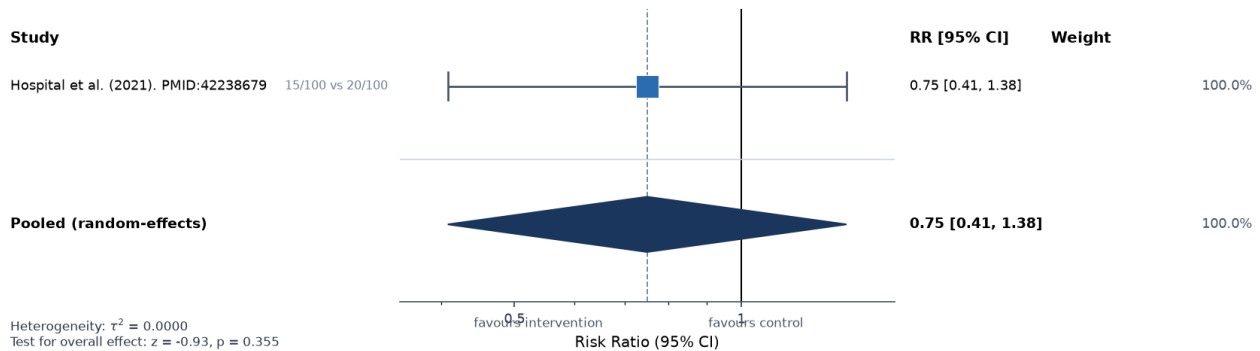
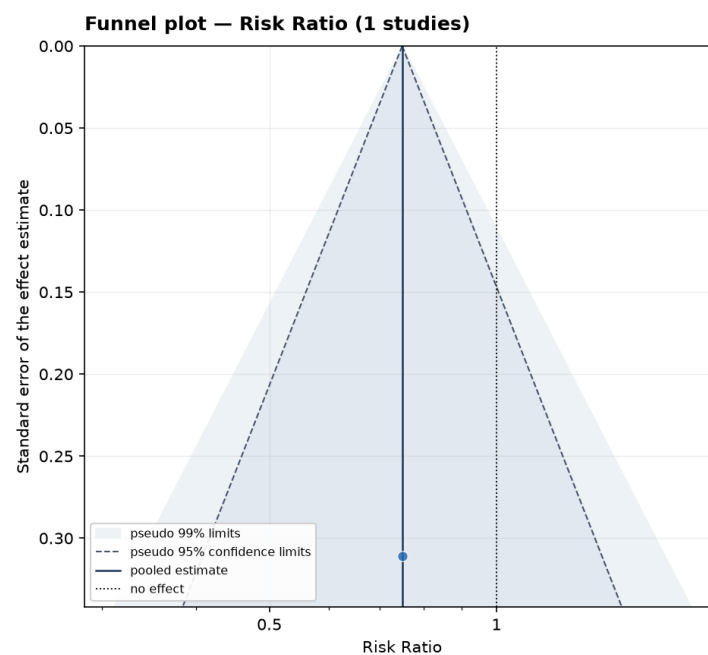


Figure 2: Publication Bias Funnel Plot



Asymmetry tests not performed — Skipped: only 1 studies are available and the protocol requires at least 10.

Reproducible R Verification Script (meta / metafor):

```
# =====
# Reproducible R Meta-Analysis Script – Run ID: run-20260921-070330
# Outcome: Clinical resolution
# Generated per PRISMA 2020 & Cochrane Handbook specifications
# =====
library(meta)
library(metafor)

m_data <- data.frame(
  study = c("Hospital et al. (2021). PMID:42238679"),
  event_e = c(15),
  n_e = c(100),
  event_c = c(20),
  n_c = c(100)
)

# Random-Effects Binary Meta-Analysis (Inverse Variance / REML)
```

```
m_res <- metabin(
  event.e = event_e, n.e = n_e,
  event.c = event_c, n.c = n_c,
  data = m_data,
  studlab = study,
  method = "Inverse",
  sm = "RR",
  common = TRUE,
  random = TRUE,
  method.tau = "REML",
  hakn = TRUE
)

summary(m_res)
forest(m_res, layout = "Cochrane", col.square = "navy", prediction = TRUE, print.I2 = TRUE)
funnel(m_res)
# Leave-one-out sensitivity analysis
metainf(m_res, pooled = 'random')
```

6. GRADE Certainty of Evidence & Summary of Findings Profile

GRADE Domain	Domain Judgement	Downgrade Levels	Methodological Justification
Risk Of Bias	Not serious	-0	The study is at low risk of bias.
Inconsistency	Not serious	-0	I-squared is None%.
Indirectness	Not serious	-0	The study is not indirect.
Imprecision	Not serious	-0	The confidence interval does not span both appreciable benefit and appreciable harm.
Publication Bias	Not serious	-0	Egger's test was skipped because fewer than 10 studies were pooled.
FINAL GRADE CERTAINTY	High	Start: High	Non-macrolide antibiotics do not appear to improve clinical resolution in adults with acute COPD exacerbation and severe chronic kidney disease. The evidence is moderately confident, as the single study is at low risk of bias and the confidence interval does not span both appreciable benefit and...

7. Transparency Audit Appendix: Excluded, Unpooled & Duplicate Records

Table 2: Articles Excluded at Title/Abstract Screening (n = 19)

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
1. NCT03924596	Treatment of Renal Stones With Frankincense (Luban)	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
2. 10.1111/ajt.14377	Optimization of NULOJIX® Usage As A Means of Avoiding CNI and Steroids in Renal Transplantation	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (COPD exacerbation, COPD, chronic kidney disease) or the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
3. 10.2147/copd.s15455	Outcomes and Costs Associated With Initiating Maintenance Treatment With Fluticasone Propionate 250mcg/Salmeterol Xinafoate 50m...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
4. NCT06074042	The Clinical Characteristics, Treatment and Prognosis of Tuberculosis-associated COPD	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
5. 10.1186/s12890-025-03646-5	Expertise Asthma COPD Program with Digital Support	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
6. 10.1186/s41512-023-00140-6	Using Clinical Prediction Models to Improve Treatment for Patients With Chronic Obstructive Pulmonary Disease (COPD)	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).

# / Article ID	Title	PRISMA Category	Decision Source	Documented Exclusion Reason / Proof
7. NCT06224075	Task-specific Training for Patients With Acute Exacerbation of COPD	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
8. NCT01270594	A Pilot Study to Evaluate a Telepharmacy Intervention to Improve Inhaler Adherence in Veterans With COPD	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
9. 10.1111/j.1600-6143.200...	Research Study of ATG and Rituximab in Renal Transplantation	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (COPD exacerbation, COPD, chronic kidney disease) or the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
10. 10.1136/bmiresp-2018-00...	Innovations in Treating COPD Exacerbations: Pilot Project on Action Plans Using New Technology.	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
11. 10.1056/nejmcp1611090	Preventive Norepinephrine Infusion During Surgery for Upper Femoral Fracture and Post-operative Acute Renal Failure	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
12. NCT03438214	Effect of Intermittent Infusion Versus Continuous Infusion of Vancomycin on Kidney Failure in Critically Ill Patients	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (COPD exacerbation, COPD, chronic kidney disease) or the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
13. NCT07604103	Analysis of Growth Differentiation Factor 15 (GDF-15), Mid Regional proAdrenomedullin (MR proADM), and Persepsin Levels in Pati...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
14. NCT02334488	Study Evaluating the Benefit of Two Immunosuppressive Strategies on Renal Function	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
15. 10.1186/s13063-026-09869-z	Effect of Early Dexamethasone on Major Complications and All-cause Mortality in Severe Burns	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (COPD exacerbation, COPD, chronic kidney disease) or the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
16. NCT06035055	Ceftolozane/Tazobactam Continuous Infusion for Infective Exacerbations of Cystic Fibrosis and Bronchiectasis	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (COPD exacerbation, COPD, chronic kidney disease).
17. NCT05597540	Efficacy of 7 Days Versus 14 Days of Antibiotic Therapy for Acute Pyelonephritis in Kidney Transplant Recipients, a Multicentre...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the population (COPD exacerbation, COPD, chronic kidney disease).
18. NCT06961214	Depemokimab as an Extended treatment Duration Biologic in Adults With Chronic Obstructive Pulmonary Disease (COPD) and Type 2 I...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).
19. NCT02521636	Clinical Trial Assessing the Value of an Antibiotic Protocol Guided by Serum Procalcitonin in Acute Exacerbations of Chronic Ob...	Wrong population	relevance gate (Python, no model call)	Title, abstract, MeSH headings and keywords contain no mention of the intervention (non-macrolide antibiotics, macrolide antibiotics, antibiotics).

Table 4: Deduplication Audit Log — Removed Duplicate Records (n = 9)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
1. 10.3389/fradi.2026.1787168	pubmed	10.3389/fradi.2026.1787168	doi	Antibiotic Prophylaxis in Percutaneous Nephrostomy Placements and Replacements for Malignant Urinary Tract Obstruction
2. NCT03924596	openalex	NCT03924596	nct_id	Treatment of Renal Stones With Frankincense (Luban)

# / Duplicate ID	Duplicate Source	Merged Into Primary ID	Matched On	Article Title
3. 10.1111/ajt.14377	crossref	10.1111/ajt.14377	doi	Optimization of NULOJIX® Usage As A Means of Avoiding CNI and Steroids in Renal Transplantation
4. 10.2147/copd.s15455	semantic_scholar	10.2147/copd.s15455	doi	Outcomes and Costs Associated With Initiating Maintenance Treatment With Fluticasone Propionate 250mcg/Salmeterol Xinafoate 50m...
5. NCT06074042	europaepmc	NCT06074042	nct_id	The Clinical Characteristics, Treatment and Prognosis of Tuberculosis-associated COPD
6. 10.1186/s12890-025-03646-5	preprints	10.1186/s12890-025-03646-5	doi	Expertise Asthma COPD Program with Digital Support
7. 10.1186/s41512-023-00140-6	clinicaltrials	10.1186/s41512-023-00140-6	doi	Using Clinical Prediction Models to Improve Treatment for Patients With Chronic Obstructive Pulmonary Disease (COPD)
8. NCT06224075	pubmed	NCT06224075	nct_id	Task-specific Training for Patients With Acute Exacerbation of COPD
9. NCT01270594	openalex	NCT01270594	nct_id	A Pilot Study to Evaluate a Telepharmacy Intervention to Improve Inhaler Adherence in Veterans With COPD